EXPERIMENT-14

Aim: Write a code using RAW sockets to implement packet sniffing.

```
from scapy.all import sniff
from scapy.layers.inet import IP, TCP, UDP, ICMP
def packet_callback(packet):
    if IP in packet:
        ip_layer = packet[IP]
        protocol = ip_layer.proto
        src_ip = ip_layer.src
        dst_ip = ip_layer.dst
        protocol_name = ""
        if protocol == 1:
            protocol_name = 'ICMP'
        elif protocol == 6:
            protocol_name = "TCP"
        elif protocol == 17:
            protocol_name = "UDP"
        else:
            protocol_name = "Unknown Protocol"
        print(f"Protocol: {protocol_name}")
        print(f"Source IP: {src_ip}")
        print(f"Destination IP: {dst_ip}")
        print("-" * 50)
sniff(iface = "Wi-fi", prn = packet_callback, filter = "ip",
      store = 0)
```

OUTPUT:

PROTOCOL : TCP
SOURCE IP : 57.155.141.117
DESTINATION IP : 192.168.1.6

PROTOCOL : UDP
SOURCE IP : 57.155.141.117
DESTINATION IP : 192.168.1.1

PROTOCOL : UDP
SOURCE IP : 57.155.141.117
DESTINATION IP : 192.168.1.1

PROTOCOL : UDP
SOURCE IP : 57.155.141.117
DESTINATION IP : 192.168.1.1

RESULT:

Thus, the program has been executed.